

TOPICS:

- 🔗 What is AI?
- 🔗 What is Machine Learning?
- 🔗 What is Deep Learning?
- 🔗 AI vs ML vs DL
- 🔗 Real-world AI applications
- 🔗 One practical example
- 🔗 Key takeaways

1. What is Artificial Intelligence?

Artificial Intelligence (AI) is the field of computer science that focuses on creating systems capable of performing tasks that normally require human-like intelligence.

AI systems can perform things like:

- Understanding language
- Recognizing images
- Making predictions
- Solving problems
- Recommending things
- Making decisions
- Generating text, images, audio or code

Examples:

- ChatGPT → language generation
- Google Maps → route prediction
- Face unlock → face recognition
- Netflix/YouTube → recommendations
- Spam filters → email classification

Simple way to remember:

AI = Machines performing intelligent tasks.

2. What is Machine Learning?

Machine Learning, or **ML**, is a subset of AI.

Instead of manually programming every single rule, we give a machine **data** and allow it to learn patterns from that data.

For example, suppose you want a system that predicts whether an email is spam.

Traditional programming:

Rules + Email → Result

Machine Learning:

Training Data + Algorithm → Learned Model

Then:

New Email + Trained Model → Spam / Not Spam

So:

ML = A way of building AI systems that learn patterns from data.

3. What is Deep Learning?

Deep Learning is a **subset of Machine Learning** that uses artificial neural networks with multiple layers.

These networks can learn complicated patterns from large amounts of data.

For example:

AI

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└─ Machine Learning

|

└─ Deep Learning

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└─ Neural Networks

Deep Learning is heavily used in:

- Computer Vision
- Speech recognition
- Natural Language Processing
- Generative AI
- Image generation
- Large Language Models

For example, systems behind modern image recognition and language models rely heavily on deep-learning techniques.

4. AI vs ML vs Deep Learning

This is the **most important part of Step 01**.

Think of it as:

AI = Big umbrella

Inside AI:

ML = One approach to achieving AI

Inside ML:

Deep Learning = One powerful ML approach

Concept	Meaning	Example
AI	Machines performing intelligent tasks	Chatbot
ML	Machines learn patterns from data	Spam detection
Deep Learning	ML using multi-layer neural networks	Image recognition

The relationship is:

Artificial Intelligence



Machine Learning



Deep Learning

But don't make the common mistake of saying **"AI and ML are the same thing."** They're related, but ML is a subset of AI.

5. Real-World AI Applications

Now connect the theory to things you actually encounter.

Healthcare

AI can help analyze medical images, assist with diagnosis, and predict risks.

Finance

AI/ML can be used for:

- Fraud detection
- Credit risk prediction
- Transaction monitoring

E-commerce

Recommendation systems analyze user behavior to suggest products.

Your activity



ML model



Predicted interests



Recommended products

Computer Vision

AI can analyze images and videos.

Examples:

- Face recognition
- Object detection

- Medical image analysis
- Quality inspection

Natural Language Processing

NLP allows computers to process human language.

Examples:

- Chatbots
- Translation
- Sentiment analysis
- Text summarization
- Voice assistants

Generative AI

Generative AI creates new content rather than simply classifying existing information.

It can generate:

- Text
- Images
- Audio
- Video
- Code

This is the area where tools like ChatGPT, image generators and coding assistants come in.

6. One Example That Connects Everything

Imagine you're building an **AI system that recognizes cats and dogs from images**.

The overall goal is:

AI

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Recognize objects intelligently

You use:

Machine Learning

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Learn patterns from training images

And specifically:

Deep Learning

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Neural network learns visual features

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Image → Cat / Dog

So one project can involve all three terms.